Those who developed radiographic evidence of progression in the treated lesion and subsequently had pathologic confirmation were included. The lesions were identified on T1 post-contrast (T1c) and T2 fluid attenuated inversion recovery (T2 FLAIR) MRI. 51 T1c and 51 T2 FLAIR radiomics features were extracted using a software developed in house. An IsoSVM machine learning classifier model trained using our institutional data (JH dataset) was tested using WF dataset.

Results: Fifty-three lesions from 43 patients were included in this study, of which all had T1c and 33 had T2 FLAIR images available. Seven lesions were pure RN pathologically, while the TP group included 37 pure tumor and 9 mixed tumor and necrosis. Twenty-two patients (51%) underwent whole brain radiotherapy with a median dose of 35Gy. Median SRS marginal dose was 18Gy (range 10-22Gy). There was no statistically significant difference among baseline characteristics between RN and TP. Of the 6 T1c radiomic features from the JH dataset, 4 demonstrated the same trend in the WF dataset (neighborhood grey tone difference matrix (NGTDM) texture strength, NGTDM coarseness, grey level run length matrix (GLRLM) grey level nonuniformity, and GLRLM run percentage). NGTDM texture strength (mean RN 161.13 vs TP 70.86, $p = 0.06$), and NGTDM coarseness (mean RN 0.03 vs TP 0.02, $p = 0.07$) approached significance. Of the 4 FLAIR features, 3 showed similar trend (kurtosis, minimum, NGTDM texture strength). The original IsoSVM model trained with all 6 T1c and 4 FLAIR features showed sensitivity of 65%, specificity of 87%, and AUC of 0.81. Due to the limited number of cases with FLAIR available, we re-trained the model with JH dataset using only the 4 T1c features showing the same trend between the two datasets, resulting in sensitivity and specificity of 80% and 73%, with AUC of 0.8. This model when tested on the WF dataset showed sensitivity and specificity of 71% and 52%, with AUC of 0.62. When the model was re-trained with the 4 T1c and 3 FLAIR features trending similarly between the two datasets, the sensitivity and specificity were 50% and 100%, and the AUC improved to 0.84.

Conclusion: Radiomics profiling of post-SRS T1c and T2 FLAIR MRI showed promising trends between RN and TP in 4 T1c features across two independent datasets. Radiomics-based IsoSVM model can distinguish RN vs TP in individual patients. Statistical significance and model performance were inferior in the validation (WF) dataset due to smaller sample size and class imbalance with fewer RN cases. Further studies with larger datasets are needed to optimize and validate the model.

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MRI-Radiomic Signature For Differentiating Low Grade Glioma From Glioblastoma Peritumoral Region

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Purpose/Objective(s): Low grade glioma (LGG) and the peritumoral region (PTR) of glioblastoma (GBM) can have an indistinguishable appearance on conventional MRI sequences. The PTR is identified as areas of T2W hyperintensity/ T1W hypointensity considered as the region of interest (ROI) in this study. The ROI in LGG usually represents tumor component, while in GBM, it is a combination of infiltrative tumor and edema. We explored the utility of MRI-radiomics in distinguishing the two groups.

Materials/Methods: T1W post-contrast, T2-FLAIR, and apparent diffusion coefficient (ADC) scans were obtained using a 1.5T MR scanner with the ROI segmented manually guided by the areas of T2W hyperintensity. Images were resampled to 1x1x1 mm^3, skull stripped, rigidly registered, and bias-field corrected. A total of 3612 features, including first-order, texture, wavelets, were extracted. Three classifiers (Support Vector Machine (SVM), K-Nearest Neighbors and Linear Discriminant Analysis), and three feature selection algorithms (Univariate T-test, Minimum Redundancy Maximum Relevance and Recursive Feature Elimination (RFE)) were applied to the extracted features. Nested 3-fold cross-validation was used to evaluate model performance using the best 10 selected features.

Abstract 3684; Table Classifier results using support vector machine-recursive feature elimination

Sequences	Sensitivity	Specificity	Accuracy	F1-score	Area Under Curve
T1W-post contrast	78%	78%	78%	78%	0.86
T2-FLAIR	88%	88%	88%	88%	0.95
ADC	93%	94%	93%	93%	0.99
All sequences	93%	94%	93%	93%	0.98

Results: A total of 73 patients (41 IDH-wild GBM and 32 grade 2 LGG) were included in this study. The best results were obtained using SVM-RFE with sensitivity, specificity, accuracy, F1-score, area under curve being 93%, 94%, 93%, 93%, 0.98, respectively. On analysis of features from individual T1, T2, and ADC sequences, the accuracy was 78%, 88%, and 93%, respectively, suggesting ADC can provide the most crucial information regarding tissue characteristics and organization.

Conclusion: This study suggests MRI-radiomics can differentiate the ROI between GBM and LGG with high accuracy. Future research can be undertaken to apply the radiomic signatures differentiating tumor from a combination of edema and infiltrative tumor that can be used in mapping target areas in radiotherapy for LGG and GBM.

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Single Center Results With SRS And FSRT In Uveal Melanoma

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Purpose/Objective(s): To assess the results of stereotactic radiosurgery (SRS) and fractionated stereotactic radiotherapy (FSRT) in patients with uveal melanoma.

Materials/Methods: Data of 417 patients treated with SRS/FSRT between September 2007 and November 2019 in our center were retrospectively evaluated. All treatments were applied via CyberKnife®.

Results: Median age of patients was 56 years. 201 (48%) patients were female and 216 (52%) were male. One patient had bilateral tumors, so the number of treated tumors was 418. 77% of tumors were T2-3, and 13% had ciliary body invasion. Total SRS/FSRT dose was <45 Gy in 100 (25%), and ≥45 Gy in 313 (75%) patients, respectively. Median tumor basal diameter was 8 mm (1.5-23 mm) and depth 8 mm (1-21 mm). Median tumor volume was 594 mm^3 (23-9.158 mm^3). Visual acuity was measured in 298 patients at the time of diagnosis. Other tumor and treatment characteristics are given in Table 1. Median follow-up was 48 months (0-263 months). The rate of 2-, 5-, and 10-year overall survival (OS) was 95%, 83%, and 68%, and distant metastasis-free survival (DMFS) was 91%, 77%, and 63%, respectively. Enucleation was performed in 92 patients; due to tumor progression in 59, and RT complication in 21 patients, respectively. The 2-, 5-, and 10-year enucleation-free survival (EFS) in all patients was 80%, 61%, and 50%, respectively. In univariate analysis, the